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Credit Risk Scoring

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INTERIM REPORT

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Table of Contents

1. Problem Definition and Real-World Context	3
2. Dataset Structure	3
2.1 Variable Types (Features).....	3
2.2 Data Size	3
2.3 Target Variable	3
2.4 Notable Characteristics	4
3. Existing Approaches in the Literature	4
4. Data and Modeling Challenges.....	5
4.1 Data Understanding and Exploratory Analysis	5
4.2 Missing Values	5
4.3 Class Imbalance	5
4.4 Categorical Variable Encoding.....	5
4.5 Multicollinearity and Feature Engineering.....	6
5. Possible Solutions or Strategies.....	6
5.1 Addressing Class Imbalance.....	6
5.2 Handling Missing Values.....	6
5.3 Feature Encoding Strategies	6
5.4 Model Selection and Optimization	6
5.5 Model Interpretability	6
References	7

1. Problem Definition and Real-World Context

Credit risk assessment is a **binary classification** problem that aims to predict whether a loan applicant constitutes a **"good"** (repays the loan) or **"bad"** (defaults on the loan) credit risk, based on financial and demographic characteristics such as account status, loan term, loan amount, and age [1].

In the banking and financial sector, accurately quantifying this risk is of critical importance: loan defaults can generate substantial revenue losses and threaten the operational stability of lending institutions [1]. Traditional credit evaluation relied heavily on manual judgment and rule-based systems, which are susceptible to human bias and inconsistency. Machine learning methods offer data-driven, objective, and scalable alternatives that can process large volumes of applicant information to support more reliable lending decisions [2]. By analysing customer attributes — such as age, employment status, savings account balance, credit amount, and loan duration — financial institutions can better understand the behavioural patterns associated with loan repayment and develop more effective risk management policies [1], [3].

2. Dataset Structure

The dataset used in this project is the **German Credit Risk Dataset**, a widely used benchmark in the credit scoring literature [3], [9], publicly available at Kaggle [13]. It contains **1,000 samples** and **9 feature variables**.

2.1 Variable Types (Features)

The dataset includes a mix of numerical and categorical predictors, as summarised in Table 1.

S/N	Feature	Description	Data Type
1	Checking_account	Status of existing checking account	Categorical
2	Duration	Duration of the credit in months	Numerical
3	Purpose	Purpose of the credit	Categorical
4	Credit_amount	Credit amount requested by the applicant	Numerical
5	Saving_accounts	Savings account balance	Categorical
6	Age	Age of the applicant in years	Numerical
7	Sex	Gender of the applicant	Categorical
8	Housing	Housing status (own / rent / free)	Categorical
9	Job	Applicant's job category	Categorical

Table 1. Features used in the German Credit Risk Dataset.

2.2 Data Size

The dataset contains exactly **1,000 samples** (rows), each representing an individual loan applicant.

2.3 Target Variable

The **Risk** variable serves as the target, making the problem suitable for supervised binary classification methods:

- Number of **good credit** samples: 700 (70%)
- Number of **bad credit** samples: 300 (30%)

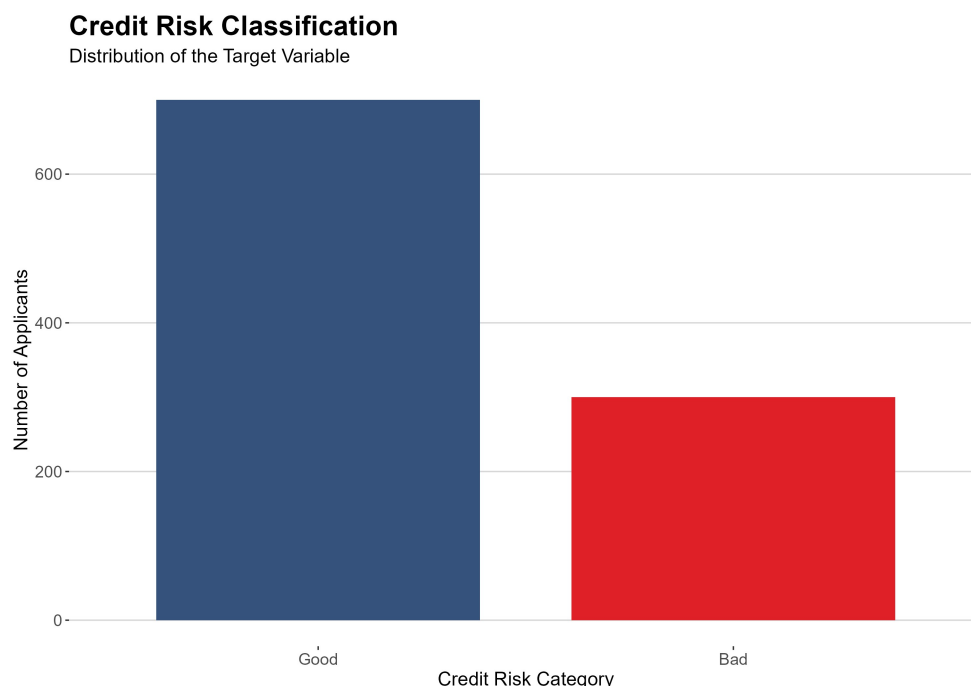


Figure 1. Distribution of the target variable (Risk): 700 good-credit vs. 300 bad-credit applicants.

2.4 Notable Characteristics

A critical feature of this dataset is the high rate of missing values in two key financial variables. The **Checking_account** variable has missing (NA) values for **39.4%** of observations, while **Saving_accounts** has a missing rate of **18.3%**. In real-world banking contexts, the absence of account information may itself carry predictive signal — applicants without formal accounts may represent a systematically distinct risk profile [10]. This missing-not-at-random (MNAR) characteristic requires careful consideration during preprocessing.

3. Existing Approaches in the Literature

Credit scoring has historically been dominated by traditional statistical methods such as logistic regression, linear discriminant analysis, and expert systems [4]. While interpretable and computationally efficient, these approaches assume linear relationships and variable independence — assumptions frequently violated in real-world credit data [1]. As a result, the research community has progressively shifted towards machine learning, which can capture nonlinear relationships and complex feature interactions more effectively [3].

Louzada et al. [3] conducted a systematic review of 187 credit scoring papers published between 1992 and 2015. Their findings identify the most commonly applied techniques as Neural Networks (17.6%), hybrid methods (16.8%), ensemble methods (16.0%), Support Vector Machines — SVMs (14.3%), and Logistic Regression (10.9%).

Logistic Regression remains the standard baseline in most comparative studies owing to its interpretability and probabilistic output [4]. Decision Trees construct transparent if-then rule structures and frequently serve as building blocks for ensemble models [6]. In a direct experiment on the German Credit Dataset, Louzada et al. [3] found SVM to offer the best balance of predictive

accuracy and computational efficiency under class-imbalanced conditions. Neural Networks can model highly complex patterns but provide limited interpretability and tend to underperform under class imbalance [6]. Ensemble methods such as Random Forest and XGBoost have become the dominant category of newly proposed methods after 2012, accounting for over 21% of contributions [3], [8].

Across 30 studies using the German Credit Dataset, reported accuracy values range from approximately 72.6% to 98.5%, underscoring the strong influence of preprocessing and experimental design [3]. Confusion matrix-derived measures — Accuracy, Sensitivity, Specificity, and F1-Score — are the most frequently reported metrics (45% of reviewed papers), followed by AUC-ROC [3]. K-fold cross-validation is the preferred validation strategy in recent work [4]. Because misclassifying a bad applicant as good carries substantially greater financial cost, cost-sensitive evaluation frameworks are also explicitly recommended [5].

4. Data and Modeling Challenges

Several interconnected challenges arise when developing machine learning models for credit risk prediction on the German Credit Dataset.

4.1 Data Understanding and Exploratory Analysis

Before any modelling step, a thorough Exploratory Data Analysis (EDA) is essential. EDA enables the investigation of relationships between variables, detection of unusual observations, and assessment of data quality. Visualization tools such as histograms, boxplots, and scatter plots help reveal how features such as age, credit amount, and loan duration differ across good- and bad-credit risk groups, and support the identification of outliers or unexpected distributions [11], [12].

4.2 Missing Values

The dataset's high missing-value rates — 39.4% in `Checking_account` and 18.3% in `Saving_accounts` — present a significant preprocessing challenge. Because missingness is likely MNAR (missing not at random), the absence of data may be systematically related to an applicant's financial situation [10]. Standard imputation methods that assume data is missing at random (MAR) may therefore introduce bias into downstream models.

4.3 Class Imbalance

The dataset contains 700 good-credit and 300 bad-credit samples, yielding a 70:30 class ratio. This imbalance tends to bias model training towards the majority class, reducing the model's sensitivity for detecting high-risk borrowers. In credit scoring, a false negative (misclassifying a bad applicant as good) is financially far more costly than a false positive [5], [7]. Models trained on imbalanced data without correction therefore tend to perform poorly on the minority class despite achieving high overall accuracy.

4.4 Categorical Variable Encoding

Five of the nine features are categorical (`Checking_account`, `Purpose`, `Saving_accounts`, `Housing`, `Job`). Since most machine learning algorithms require numerical input, appropriate encoding strategies must be applied without introducing artificial ordinal relationships among nominal categories [9]. Choices such as one-hot encoding vs. ordinal encoding can meaningfully affect model performance and interpretability.

4.5 Multicollinearity and Feature Engineering

Some features may exhibit high inter-correlation, which can reduce the stability of statistical models. Effective feature engineering — including appropriate scaling, variable transformation, and construction of derived features — is essential to maximise model performance and capture underlying relationships in the data [1], [8].

5. Possible Solutions or Strategies

5.1 Addressing Class Imbalance

SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic observations for the minority class by interpolating between existing minority samples and has been widely demonstrated to improve recall for bad-credit applicants [7]. **Random undersampling** balances the dataset by reducing majority-class instances, though at the risk of discarding useful information [7]. **Cost-sensitive learning** assigns higher misclassification penalties to the minority class during training, which is especially appropriate for credit scoring where false negatives are costlier [5]. Ensemble methods such as **Balanced Random Forest** and **EasyEnsemble** incorporate imbalance correction directly into their architecture [7].

5.2 Handling Missing Values

Mean/mode imputation offers simplicity but ignores the MNAR structure of the data [3]. **Multiple imputation** provides a more robust statistical approach by accounting for uncertainty in imputed values [3]. **Model-based imputation** using k-NN or Random Forest imputers can yield more accurate estimates by leveraging patterns in observed data [7]. Alternatively, treating *"missing"* as a distinct category for nominal variables like `Checking_account` may preserve an informative signal that simple imputation would eliminate [9].

5.3 Feature Encoding Strategies

One-Hot Encoding is recommended for nominal variables such as `Purpose` and `Housing`, as it avoids imposing artificial ordinal relationships [9]. **Ordinal Encoding** is appropriate for the `Job` variable, which exhibits a natural ordering across skill levels. For high-cardinality categorical variables, **target encoding** provides a data-efficient alternative [8].

5.4 Model Selection and Optimization

Ensemble methods — particularly **Random Forest** and **XGBoost** — are recommended as strong baselines given their demonstrated performance on this problem [3], [8]. Stratified k-fold cross-validation is essential for obtaining reliable generalisation estimates on a relatively small dataset [4]. Hyperparameter tuning via grid search or Bayesian Optimization can substantially improve model performance [8].

5.5 Model Interpretability

SHAP (SHapley Additive exPlanations) provides instance-level feature contribution explanations compatible with any model architecture [6]. **LIME (Local Interpretable Model-agnostic Explanations)** generates local approximations for explaining individual predictions [6]. These tools bridge the gap between high-performance black-box models and the transparency requirements of financial decision-making.

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